Farros Hilmi Syafei

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Full-stack engineer and Informatics Engineering graduate. Ships products end to end: React/Next.js frontends, Python (FastAPI) and TypeScript backends, PostgreSQL, Redis and Kafka, deployed with Docker and Vercel. Built a production document-verification API during an internship and an event-sourced platform of nearly 1,000 source files as a thesis. Works daily with AI coding agents (Claude Code), with tests and CI as the safety net.

EDUCATION

Institut Teknologi Sepuluh Nopember (ITS), Surabaya — Graduated Oct 2026

B.S. Informatics Engineering · GPA 3.72 / 4.00

EXPERIENCE

Software Engineering Intern — PT Nusantara Baskara Jaya (Nusaraya) — Dec 2024 – Jun 2025

Logistics platform · Jakarta — Python · FastAPI · PaddleOCR · YOLO · Redis · Docker

- Watched how operations staff checked driver and vehicle documents (phone photos retyped by hand), then designed and built a REST API to replace that workflow for onboarding and dispatch.
- Stateless FastAPI service with API-key auth, rate limiting, health checks and metrics. It processes KTP, SIM, STNK and KIR at 95% document-type accuracy and 83–90% field accuracy, and ships as a Docker image (GHCR).
- Designed the API contract around operations: `match / no_match / needs_review` with reason codes, so uncertain cases go to a human. Added perceptual-hash duplicate-upload detection.

Teaching Assistant, Operating Systems — Dept. of Informatics, ITS — Feb 2023 – Jun 2026

- Led lab sessions, wrote practicum problems and graded assignments with written feedback across multiple cohorts.

Full Stack Web Developer (Bootcamp) — Binar Academy — Jul 2022 – Jan 2023

- Built MyAirFare, a responsive app to search flights and book round-trip tickets.

PROJECTS

HCIE — event-sourced adaptive tutoring platform (thesis) — FastAPI · PostgreSQL · Kafka · Next.js · React

- Full-stack learning platform: FastAPI backend with a transactional outbox to Kafka feeding CQRS read models, and a Next.js/React frontend using TanStack Query and Recharts dashboards. Runs with Docker Compose and GitHub Actions CI.
- Personalizes for new learners without any interaction history (cold-start AUC 0.6051 vs. BKT 0.5963, on 1.98M interactions from 232k learners). State rebuilds from the event log with bit-identical replay.

Lyceum — reproducibility index for ML research — TypeScript · TanStack Start · Bun · PostgreSQL · Typesense · Valkey

- Monorepo with an indexer (arXiv and Papers with Code archive), cron jobs that check papers' repositories and datasets, a React web app with search and paper pages, and an MCP server for AI agents. PostgreSQL behind PgBouncer, with Typesense for search.

Portfolio with live OCR demo — farroshsyportfolio.vercel.app — Next.js 16 · React 19 · TypeScript · Tailwind v4 · Vercel

- Content-driven site with a live document-verification demo. A server-side route proxies uploads to the OCR API, keeping the API key server-side, enforcing size, type and document-type limits, and storing nothing.

dokumen-id — dokumen-id.vercel.app — Next.js · TypeScript

- Public tool that decodes Indonesian NIK numbers entirely in the browser, with no network requests, so personal data never leaves the device. A Swift package version (DokumenKit) covers iOS.

rota — self-hostable on-call alerting — Rust · Docker

- Single-binary service that receives alerts, runs health checks, escalates unacknowledged alerts on a schedule and survives restarts.

SKILLS

Frontend	React, Next.js (App Router), TypeScript, TanStack Query/Start, Tailwind CSS, Recharts
Backend	Python (FastAPI), TypeScript (Bun, Node), Rust, REST API design, event sourcing / CQRS, MCP
Data & Infra	PostgreSQL, Redis/Valkey, Kafka/Redpanda, Typesense, Docker / Compose, GitHub Actions, Vercel
Also	Swift/SwiftUI (iOS), PaddleOCR, YOLO, Java, C/C++, SQL; AI-assisted development (Claude Code)
Languages (spoken)	Indonesian (native), English (fluent)